

MERSA MATRUH, Egypt

WMO: 623060

Lat: 31.325N

Lon: 27.222E

Elev: 94

StdP: 14.65

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.0	47.8	28.6	22.7	64.3	33.0	27.6	59.9	33.2	58.4	30.0	57.0	10.7	240	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.9	68.4	88.2	72.3	86.4	73.5	78.0	84.8	77.1	84.1	76.2	83.4	14.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	134.5	82.4	75.0	131.5	82.1	73.7	125.8	81.5	41.6	84.6	40.6	84.2	39.8	83.3	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 27.4	23.2	20.7	DB	42.2	105.7	2.1	4.2	40.7	108.8	39.5	111.3	38.3	113.6	36.7	116.7	(4)
(5)			WB	38.2	80.1	2.2	1.6	36.6	81.3	35.3	82.2	34.1	83.1	32.5	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	57.0	57.9	60.7	64.9	70.0	75.2	79.0	80.0	78.3	73.4	66.4	60.0		
	DBStd	9.00	3.28	4.20	4.74	5.10	4.83	3.56	2.35	1.88	3.05	3.55	3.76	3.29		
	HDD50	1	0	1	0	0	0	0	0	0	0	0	0	0		
	HDD65	855	249	203	155	58	6	0	0	0	0	0	26	158		
	CDD50	6801	218	224	331	447	619	756	899	931	848	725	493	310		
	CDD65	2179	1	6	21	55	159	306	434	466	398	261	68	4		
	CDH74	18209	12	56	196	439	941	2197	4159	4891	3573	1494	236	15		
(13)	CDH80	5105	1	14	75	170	339	510	1179	1544	967	277	28	1		
(14)	Wind	WSAvg	10.6	11.4	11.9	11.8	11.6	10.5	10.7	11.2	9.9	9.3	9.1	10.9		
(15) Precipitation	PrecAvg	4.8	1.3	0.7	0.4	0.1	0.1	0.0	0.0	0.0	0.0	0.5	0.7	1.2		
	PrecMax	8.7	5.0	2.8	1.3	0.6	0.9	0.0	0.0	0.7	1.2	3.5	4.1	4.8		
	PrecMin	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.6	1.1	0.7	0.3	0.2	0.2	0.0	0.0	0.1	0.2	0.7	0.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.1	80.5	89.4	93.6	97.1	98.3	93.3	91.8	98.6	91.6	83.9	75.5		
		MCWB	56.6	56.9	61.1	63.5	66.9	67.7	73.3	74.0	70.6	67.6	63.0	57.9		
	2%	DB	68.4	73.4	78.8	84.4	88.2	87.8	88.1	88.1	89.6	85.8	79.1	71.5		
		MCWB	55.6	55.5	58.1	60.8	65.1	70.3	74.7	76.1	72.2	67.5	64.3	58.6		
	5%	DB	66.2	68.4	73.0	78.5	82.5	84.3	86.3	87.0	86.1	82.5	76.7	69.4		
		MCWB	55.8	55.3	57.6	60.9	64.8	71.8	74.7	75.9	73.5	69.8	64.3	58.5		
	10%	DB	64.4	65.6	68.3	73.5	78.4	82.3	84.9	86.0	84.3	80.5	74.2	67.6		
		MCWB	55.2	55.5	57.9	60.5	65.5	71.7	74.5	75.6	73.3	69.3	63.5	57.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	61.5	63.7	67.1	71.1	76.7	78.6	79.5	78.3	74.8	71.4	64.6		
		MCDB	67.4	68.6	75.8	81.2	79.3	83.6	85.4	85.8	85.3	81.1	76.7	70.6		
	2%	WB	58.6	59.4	61.6	64.8	69.3	75.2	77.4	78.4	76.7	73.0	68.2	61.7		
		MCDB	65.1	66.8	70.5	75.5	77.9	81.8	84.2	84.8	83.9	80.1	75.0	67.9		
	5%	WB	57.3	57.9	60.2	63.3	68.3	74.1	76.5	77.5	75.5	71.6	66.3	60.1		
		MCDB	64.0	65.1	68.1	72.3	76.5	80.7	83.4	84.4	83.0	79.1	73.7	66.7		
	10%	WB	56.1	56.6	59.0	62.1	67.3	72.9	75.6	76.7	74.4	70.2	64.7	58.8		
		MCDB	62.9	63.8	66.7	70.4	75.2	79.7	82.8	83.9	82.2	78.0	72.4	65.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	15.2	15.8	16.3	16.1	14.2	13.0	12.9	13.8	14.4	15.2	14.5		
		MCDBR	18.0	20.9	23.8	26.3	25.4	18.3	15.0	14.2	17.4	17.9	19.1	17.5		
		MCWBR	10.3	10.4	10.7	10.9	10.1	8.5	6.9	6.6	8.0	8.6	9.7	9.9		
		MCDBR	15.8	16.9	18.9	19.4	17.4	14.3	13.8	13.2	15.0	14.9	15.9	14.9		
	5% WB	MCWBR	10.0	10.1	10.3	9.8	8.6	7.7	6.9	6.7	8.2	8.4	9.7	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.385	0.434	0.465	0.509	0.500	0.437	0.419	0.423	0.446	0.425	0.393	0.386		
	taud		2.204	2.034	1.947	1.860	1.916	2.134	2.194	2.191	2.125	2.187	2.237	2.218		
	Ebn at Noon		260	257	258	252	255	270	275	272	261	257	256	252		
	Edh at Noon		37	49	57	65	62	49	46	46	47	41	36	35		
(44) All-Sky Solar Radiation	RadAvg		960	1211	1627	1974	2192	2371	2317	2125	1834	1417	1052	866		
	RadStd		47	77	67	71	74	52	41	36	46	51	45	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	+257	+153
(47) Regional (1 neighbor)		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	+257	+153

Nomenclature: See separate page